Cross-Border E-Commerce Supply Chain Risk Classification with Explainable AI and Structured Decision Support

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Abstract. Cross-border e-commerce supply chains are affected by interconnected operational, geographic, economic, and logistics factors. Delayed deliveries can increase business costs, reduce customer satisfaction, and make supply chain planning more difficult. This capstone project analyzes a public cross-border e-commerce supply chain dataset containing 25,000 records and develops machine learning classification models to predict whether an order is likely to be delivered on time or delayed.
The project includes data validation, data cleaning, exploratory data analysis, feature selection, preprocessing, model training, model evaluation, threshold optimization, and explainable artificial intelligence. Four classification approaches were evaluated: a Dummy Classifier, Logistic Regression, Random Forest, and Gradient Boosting. Logistic Regression was selected as the final model because it provided the most practical balance between identifying delayed shipments, recognizing on-time shipments, and supporting model interpretation.
A business-oriented probability threshold of 0.37 was selected using validation data to give greater importance to detecting delayed shipments. On the unseen testing dataset, the final model achieved 71.14% accuracy, 84.01% delayed-delivery precision, 77.28% delayed-delivery recall, and a delayed-class F1-score of 0.8051. Logistic Regression coefficient analysis and SHAP explainability both identified port congestion as the most influential predictor of delivery-delay risk.
The resulting workflow provides an interpretable foundation for early delivery-risk identification and future structured decision-support development in cross-border e-commerce supply chains.

Keywords: Cross-Border E-Commerce · Supply Chain Analytics · Delivery Risk Prediction · Machine Learning · Explainable AI · Classification Models

1 Introduction

Cross-border e-commerce logistics is an important area within supply chain analytics because international deliveries are influenced by many connected operational, geographic, economic, and transportation factors. These factors include shipping mode, supplier reliability, port congestion, tariffs, fuel costs, inventory levels, customer location, warehouse conditions, and market demand. Since e-commerce customers expect fast and reliable delivery, shipment delays can reduce customer satisfaction, increase operational costs, and make supply chain planning more difficult.

This project uses a public cross-border e-commerce supply chain dataset obtained from Kaggle. The dataset contains shipment, product, customer, logistics, inventory, cost, and delivery-related attributes. A public dataset was selected so that the analysis would not involve confidential company, customer, vendor, or employer-specific information.

The main data problem addressed in this project is whether supply chain and logistics factors can be used to predict whether a cross-border e-commerce order will be delivered on time or delayed. This problem is important because delivery performance is rarely influenced by a single variable. Instead, it may depend on a combination of logistics conditions, geographic factors, market characteristics, supplier performance, inventory availability, and external trade-related risks.

The project develops and evaluates machine learning classification models for delivery-delay prediction. In addition to generating predictions, the analysis examines which factors contribute most strongly to model outcomes using Logistic Regression coefficients and SHAP explainability. The final goal is to provide an interpretable predictive workflow that can support earlier identification of delivery risk and future supply chain decision-support development.

1.1 Goals of this Research

The main goal of this project is to develop an interpretable machine learning workflow that predicts delivery-delay risk in cross-border e-commerce supply chains. The specific goals are:

- Validate and clean the selected dataset while preserving the original records.
- Explore delivery-performance patterns across product, geographic, supplier, inventory, warehouse, economic, and logistics variables.
- Review candidate predictors for identifiers, redundancy, timing uncertainty, and possible data leakage.
- Build and compare classification approaches for predicting whether a shipment will be delayed or delivered on time.
- Address target-class imbalance using stratified data splitting, class weighting, and appropriate evaluation metrics.
- Evaluate model performance using accuracy, balanced accuracy, precision, recall, F1-score, F2-score, ROC-AUC, PR-AUC, classification reports, and confusion matrices.
- Select a practical decision threshold that gives greater importance to detecting delayed shipments.
- Explain model predictions using Logistic Regression coefficients and SHAP analysis.

– Save the final preprocessing transformer, trained model, decision threshold, and feature information as a reusable model package.
– Translate the predictive results into business-friendly insights and a foundation for future structured decision support.

2 Data

The project uses the public Cross-Border E-Commerce Supply Chain Dataset from Kaggle. The dataset contains 25,000 records and 43 original attributes related to products, customers, markets, suppliers, inventory, warehouses, logistics, costs, external conditions, and delivery outcomes.

The original target variable is `Delivery_Time_OnTime`. Based on the attribute name, a value of 1 represents an on-time delivery and a value of 0 represents a delayed or not-on-time delivery. For predictive modeling, a new target called `Delivery_Delayed` was created so that 1 represents a delayed delivery and 0 represents an on-time delivery. This allowed the positive class to represent the main business risk addressed by the project.

The dataset is suitable for this analysis because it contains several operational and external-risk variables that may be associated with delivery performance. These include shipping mode, supplier reliability, port congestion, tariff rate, fuel cost index, warehouse capacity, inventory level, distance to customer, shipping cost, product characteristics, country, and region.

Only public dataset information was used. No private company, customer, vendor, or employer-specific data was included in the analysis.

Table 1. Dataset and Predictive Modeling Components

Component	Description
Domain	Cross-border e-commerce supply chain analytics
Data Source	Cross-Border E-Commerce Supply Chain Dataset from Kaggle
Dataset Size	25,000 records and 43 original attributes
Original Target	`Delivery_Time_OnTime`, where 1 represents on-time delivery and 0 represents delayed delivery
Modeling Target	`Delivery_Delayed`, where 1 represents delayed delivery and 0 represents on-time delivery
Final Predictor Set	32 predictors consisting of 25 numerical and 7 categorical variables
Modeling Task	Binary classification of delayed and on-time shipments
Models Evaluated	Dummy Classifier, Logistic Regression, Random Forest, and Gradient Boosting
Final Model	Logistic Regression with a business-oriented threshold of 0.37
Expected Use	Delivery-risk prediction, explainability, and future structured decision-support development

Table 1 summarizes the dataset and the main predictive-modeling components used in this project.

3 Data Collection and Extraction

3.1 Data Source

The data source used for this project is the Cross-Border E-Commerce Supply Chain Dataset from Kaggle [1]. It is a public dataset representing cross-border e-commerce supply chain operations and contains 25,000 records with 43 original attributes. The attributes cover products, customers, markets, suppliers, warehouses, inventory, transportation, costs, external conditions, and delivery outcomes.

The dataset was selected because it supports the project objective of analyzing delivery risk and developing a binary classification model to predict whether an order is likely to be delayed or delivered on time. Only the publicly available dataset was used, and the project does not contain confidential company, customer, vendor, or employer-specific information.

3.2 Dataset Format

The dataset was downloaded from Kaggle in a structured CSV format. The CSV file was loaded into Python using the pandas library for inspection, cleaning, exploratory analysis, and predictive modeling. In the tabular dataset, each row represents a cross-border e-commerce order or supply chain transaction, while each column represents an attribute related to the product, customer, supplier, warehouse, logistics process, cost, external conditions, or delivery outcome.

The CSV format was appropriate for this project because it supports efficient loading, validation, transformation, visualization, and machine learning analysis in Python. The original file was preserved, and a separate processed version was created for later stages of the project.

3.3 Data Gathering and Scraping Techniques

No web scraping was required because the dataset was publicly available on Kaggle as a downloadable file. The data-gathering process consisted of accessing the dataset page, downloading the dataset package, extracting the CSV file, and loading it into Python for validation and analysis.

The downloaded dataset was stored in the project directory, while the original file was preserved without modification. A separate working copy was used for cleaning, exploratory analysis, feature preparation, and predictive modeling. Since the project relied only on a public dataset, no private company, customer, vendor, or employer-specific information was collected.

3.4 Major Data Attributes

The dataset contains 43 original attributes covering several areas of cross-border e-commerce supply chain operations. These include product and market variables, customer and geographic variables, supplier and warehouse variables, inventory measures, logistics and transportation variables, external-risk indicators, cost-related variables, and delivery outcomes.

The original delivery-status attribute is `Delivery_Time_OnTime`, where a value of 1 represents an on-time delivery and a value of 0 represents a delayed or not-on-time delivery. For model development, this outcome was converted into `Delivery_Delayed`, where 1 represents a delayed delivery and 0 represents an on-time delivery.

Important candidate predictors included `Shipping_Mode`, `Supplier_Reliability`, `Port_Congestion_Index`, `Tariff_Rate`, `Fuel_Cost_Index`, `Warehouse_Capacity`, `Inventory_Level`, `Distance_to_Customer`, `Shipping_Cost`, `Product_Price`, `Country`, and `Region`. These variables were relevant because delivery risk may be associated with transportation conditions, supplier performance, inventory availability, geographic factors, external trade conditions, warehouse capacity, and logistics costs.

The attribute `Lead_Time_Days` showed a strong relationship with delivery performance during exploratory analysis. However, it was excluded from the final predictor set because its timing was uncertain and it could represent information that becomes available only after, or very close to, the delivery outcome. This conservative decision reduced the risk of data leakage.

3.5 Additional Extraction Details

After downloading the dataset, the CSV file was loaded into Python using the pandas library. The initial extraction and validation process included checking the number of rows and columns, reviewing column names, inspecting data types, examining sample records, identifying missing values, checking duplicate rows, and confirming the availability of the delivery-status target and important candidate predictors.

The original dataset contained 25,000 records and 43 attributes. After validation and cleaning, the processed dataset was saved as `cross_border_ecommerce_supply_chain_cleaned.csv` in the `data/processed` directory. This cleaned file was then used for exploratory data analysis, feature selection, preprocessing, machine learning classification, threshold evaluation, explainability analysis, and final model packaging.

4 Data Cleaning and Curation

4.1 Data Curation Process

The raw cross-border e-commerce supply chain dataset was preserved in its original form, and a separate working copy was created before making any changes.

This ensured that the original dataset remained unchanged and could be compared with the processed version when needed.

The data curation process began by reviewing the dataset dimensions, column names, data types, sample records, categorical values, and numeric summary statistics. The dataset was then checked for missing values, blank strings, hidden placeholder values, duplicate rows, repeated order identifiers, inconsistent category labels, invalid numeric ranges, and incorrectly formatted dates.

The review showed that the dataset was already highly complete and consistently structured. No missing values, hidden placeholder values, duplicate rows, or repeated order identifiers were found. The main data-type issue involved the `Date` attribute, which was originally stored as text. It was converted to a datetime format to support time-based analysis. Text-based attributes were also standardized by removing leading and trailing spaces.

The major steps followed during data curation were:

1. Preserve the original dataset.
2. Create a separate working copy for cleaning.
3. Review the dataset structure and data types.
4. Check for missing values and hidden placeholders.
5. Check for duplicate rows and repeated order identifiers.
6. Review categorical and numeric attributes for inconsistencies.
7. Convert the `Date` attribute to datetime format.
8. Standardize text-based attributes.
9. Validate and save the cleaned dataset.

After the cleaning steps were completed, the dataset was validated again by checking the number of records and attributes, missing-value totals, duplicate counts, unique order identifiers, and updated data types. The cleaned dataset was then saved separately as `cross_border_ecommerce_supply_chain_cleaned.csv` in the `data/processed` directory.

4.2 Tools and Techniques

Python was used to complete the data-cleaning and preprocessing activities in a Jupyter Notebook environment. Pandas was the primary library used because it provides functions for loading, inspecting, validating, transforming, and exporting structured datasets.

The dataset was loaded using the Pandas `read_csv()` function. A separate working copy was created using `copy()` so that the original dataset would remain unchanged. The functions `head()`, `info()`, `describe()`, `nunique()`, and `value_counts()` were used to inspect sample records, data types, descriptive statistics, unique values, and category distributions.

Missing values were checked using `isna()` and column-level totals. Text attributes were also reviewed for blank strings and common placeholder values such as `NA`, `N/A`, `None`, `Null`, `NaN`, question marks, and hyphens.

Duplicate rows were checked using `duplicated()`, and the `Order_ID` attribute was reviewed separately to confirm that each order identifier was unique.

Categorical attributes were checked using unique-value counts, while numeric attributes were reviewed using minimum, maximum, mean, and median values to identify invalid or unreasonable ranges.

The `Date` attribute was converted using `to_datetime()`, and text fields were standardized using `str.strip()`. Potential outliers were not automatically removed because unusually high or low values may represent valid supply chain conditions rather than data errors.

Encoding, scaling, feature selection, and class-imbalance handling were intentionally not applied during the data-cleaning stage. These steps were later performed within the machine learning workflow so that preprocessing could be fitted using only the training data and then applied consistently to validation and testing data. This approach helped reduce the risk of data leakage.

4.3 Handling Missing Values

The dataset was checked for both standard and hidden missing values. Standard null values were evaluated across all 43 attributes. Text-based attributes were also checked for blank strings and common placeholder values, including `NA`, `N/A`, `None`, `Null`, `NaN`, question marks, and hyphens.

The assessment found no standard missing values, blank strings, or hidden missing-value placeholders. Each attribute contained 25,000 non-null observations. Therefore, no records or attributes were removed, and no mean, median, mode, forward-fill, or model-based imputation was required.

Retaining the original observed values was the most appropriate strategy because applying imputation when no values were missing could introduce artificial information and unnecessarily change the dataset's original distributions.

Although no values required imputation in the cleaned dataset, missing-value safeguards were still included in the predictive pipeline. Numerical variables used median imputation, while categorical variables used most-frequent imputation. These safeguards were fitted only on the training data so that the same preprocessing rules could be applied consistently to validation, testing, and future prediction records without introducing data leakage.

4.4 Final Dataset Size

The original dataset contained 25,000 records and 43 attributes. Because no missing, duplicate, or invalid records were identified, the cleaning process did not require the removal of any rows or columns. The final cleaned dataset therefore contains 25,000 records and 43 attributes.

4.5 Definitions of Essential Attributes

The following attributes are especially important to the delivery-risk analysis:

- `Delivery_Time_OnTime`: Binary delivery-status indicator representing the delivery outcome of an order. Based on the column name, a value of 1 is

interpreted as an on-time delivery, while 0 is interpreted as a delayed or not-on-time delivery.
- `Lead_Time_Days`: Number of days required to complete the supply and delivery process.
- `Supplier_Reliability`: Measure representing the consistency and dependability of the supplier.
- `Port_Congestion_Index`: Index representing the level of congestion affecting port operations and shipment movement.

- `Shipping_Mode`: Transportation method used to deliver the order.
- `Shipping_Cost`: Cost associated with transporting and delivering the order.
- `Inventory_Level`: Quantity of inventory available to support customer demand and order fulfillment.
- `Tariff_Rate`: Tariff percentage associated with the cross-border movement of products.

4.6 Dependent and Independent Variables

The original delivery-outcome variable is `Delivery_Time_OnTime`, where class 1 represents an on-time delivery and class 0 represents a delayed or not-on-time delivery. For predictive modeling, the target was recoded as `Delivery_Delayed`, where 1 represents a delayed delivery and 0 represents an on-time delivery. This coding makes delayed delivery the positive class because it is the main operational risk addressed by the project.

The dataset contains 19,278 delayed records and 5,722 on-time records. Delayed deliveries therefore represent approximately 77.11% of the dataset, while on-time deliveries represent 22.89%. Because the target is imbalanced, model performance was evaluated using accuracy together with balanced accuracy, precision, recall, F1-score, F2-score, confusion matrices, ROC-AUC, and PR-AUC.

A total of 32 predictors were selected for the final machine learning workflow. These consisted of 25 numerical variables and 7 categorical variables. The predictor set included time-related, product, market, geographic, supplier, warehouse, inventory, logistics, cost, and external-risk variables.

The attributes `Order_ID`, `Product_ID`, and `Date` were excluded because they function as identifiers or were represented through other time-related variables. The original target and derived target were excluded from the predictor set. Variables representing service outcomes, optimization outputs, alternative outcomes, or information that could become available only after the delivery result were also excluded to reduce data leakage.

The attribute `Lead_Time_Days` was excluded from the final predictor set because its timing was uncertain. Although it showed a strong relationship with delivery performance during exploratory analysis, using it could make the model depend on information that may not be available early enough for practical prediction.

5 Exploratory Data Analysis

5.1 Definition and Importance

Exploratory Data Analysis, or EDA, is an approach used to gain insight into a dataset, uncover its structure, identify important variables, and detect outliers or anomalies [2]. It includes examining data distributions, comparing variables, and looking for patterns or possible issues such as outliers, class imbalance, and highly related features.

EDA is important because it helps show what the data is actually telling us and provides direction for the next stages of the project. In this project, EDA helped me understand the delivery-outcome distribution, identify variables that may be related to delayed deliveries, and recognize issues that should be reviewed before modeling, such as class imbalance and possible data leakage.

5.2 EDA Techniques

Exploratory data analysis can include several techniques, such as descriptive statistics, frequency tables, distribution analysis, univariate analysis, bivariate analysis, correlation analysis, and outlier detection. Visualizations such as bar charts, histograms, boxplots, and heatmaps can also make patterns in the data easier to understand.

For this project, I used descriptive statistics to review the mean, median, standard deviation, minimum, and maximum values of the numerical variables. I used frequency tables and bar charts to study the target variable and categorical attributes such as product category, shipping mode, region, country, and customer segment. Histograms were used to examine numerical distributions, while boxplots and grouped statistics were used to compare delayed and on-time deliveries. I also used correlation analysis and a heatmap to study relationships between numerical variables. Finally, the interquartile range, or IQR, method was used to identify possible outliers in selected variables.

5.3 Techniques and Results

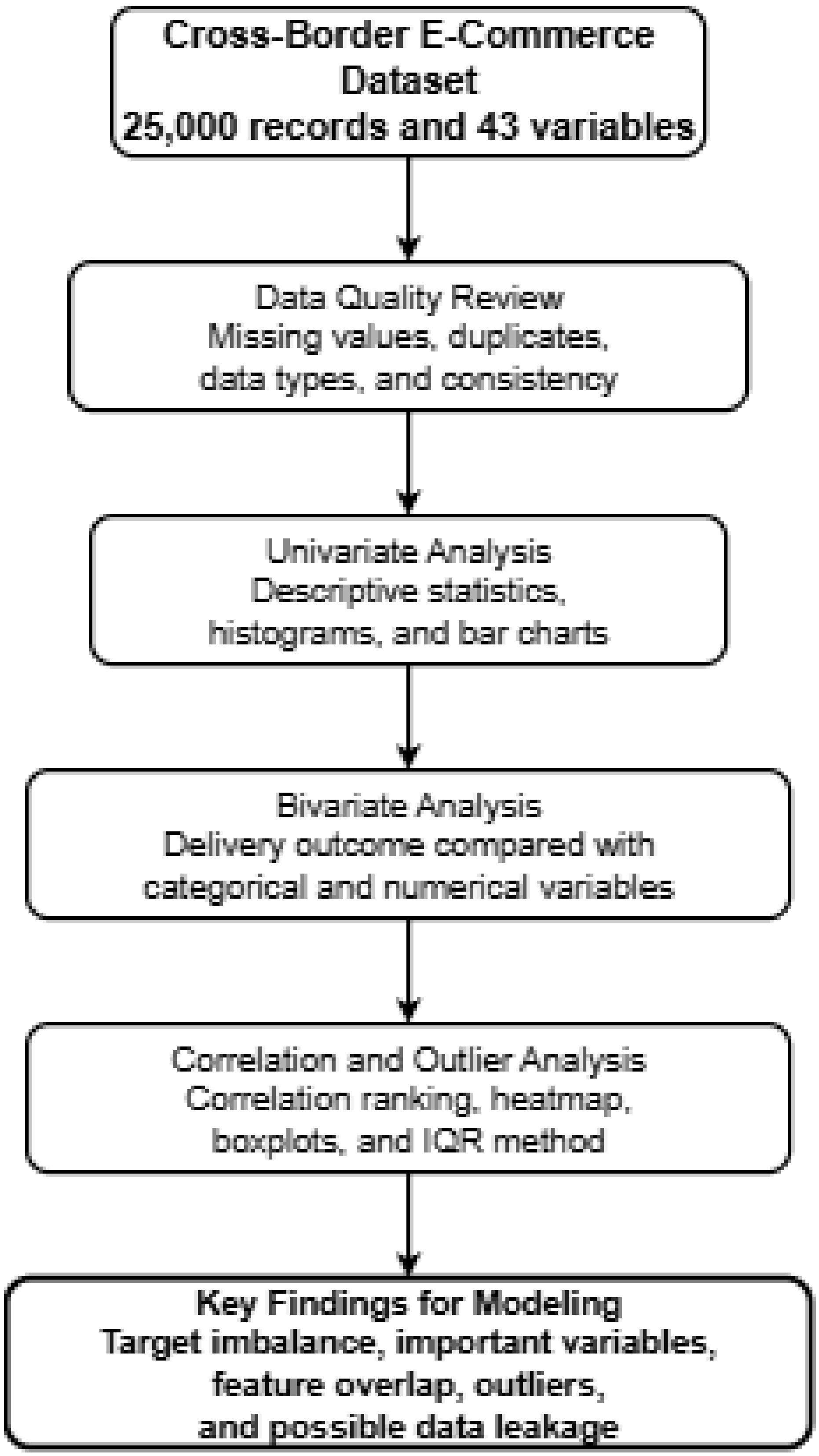

Fig. 1. Exploratory data analysis workflow used in this project.

I first used descriptive statistics and distribution plots to understand the numerical variables. The mean, median, standard deviation, minimum, and maximum values were reviewed for variables such as lead time, supplier reliability, port congestion, shipping cost, inventory level, and carbon emission. Histograms showed that lead time and supplier reliability were fairly evenly distributed, while shipping cost and carbon emission were right-skewed. The higher values were not removed automatically because they may represent real supply chain conditions.

I then used frequency tables and bar charts to study the target variable and the categorical attributes. The delivery outcome was imbalanced, with 19,278 delayed deliveries and 5,722 on-time deliveries. This means that delayed deliveries represented 77.11% of the dataset, while on-time deliveries represented 22.89%. Product categories, shipping modes, regions, and customer segments were fairly balanced. Country-level record counts showed more variation, so percentages were used when comparing delivery performance across countries.

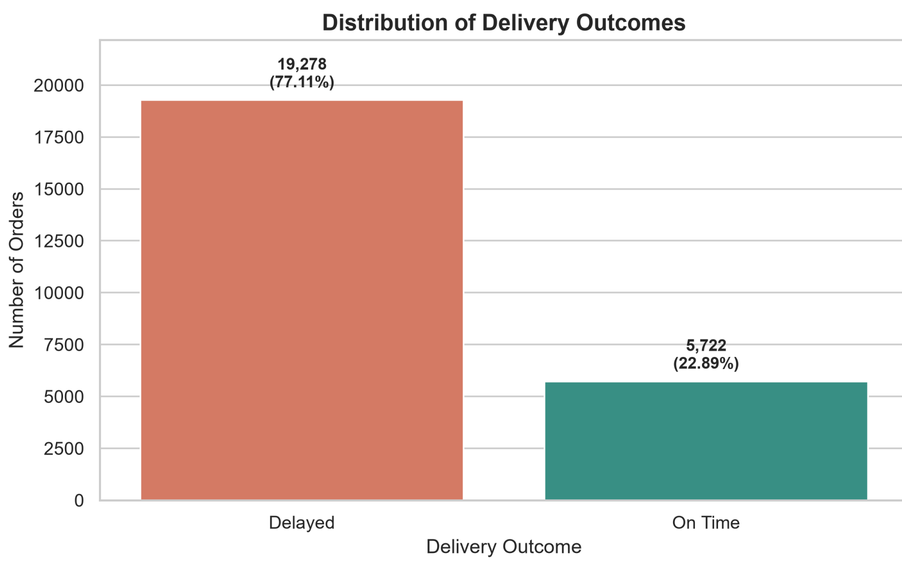

Fig. 2. Distribution of delayed and on-time deliveries.

I next compared delivery performance across shipping modes, regions, product categories, and countries. The on-time delivery percentages were very similar across shipping modes, regions, and product categories. The difference between the highest and lowest on-time rates was only about 1.32 percentage points for shipping mode, 1.14 percentage points for region, and 1.61 percentage points for product category. Country-level differences were slightly larger, but the overall range was still only about 3.14 percentage points. This suggests that these categorical variables may not explain delivery performance strongly when viewed by themselves.

Boxplots and grouped summary statistics were then used to compare numerical variables between delayed and on-time deliveries. The clearest difference was found in `Lead_Time_Days`. Delayed deliveries had an average lead time of 21.42 days, while on-time deliveries had an average of 8.54 days. A similar pattern was found for `Port_Congestion_Index`, where delayed deliveries had an average value of 55.89 compared with 29.58 for on-time deliveries. In comparison, supplier reliability and shipping cost were almost the same for both delivery groups.

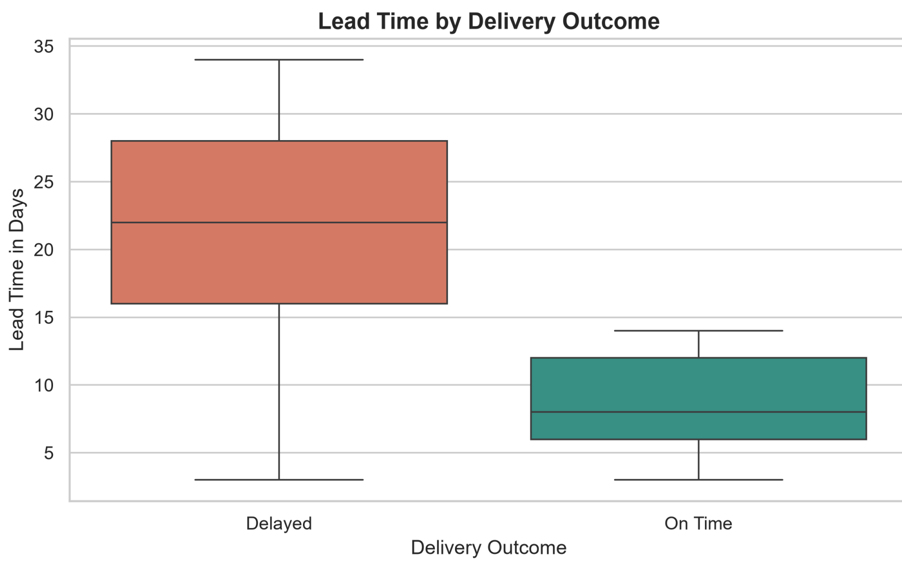

Fig. 3. Lead time comparison between delayed and on-time deliveries.

Port congestion also showed a noticeable difference between the two delivery groups. Delayed deliveries had an average `Port_Congestion_Index` value of 55.89, compared with 29.58 for on-time deliveries.

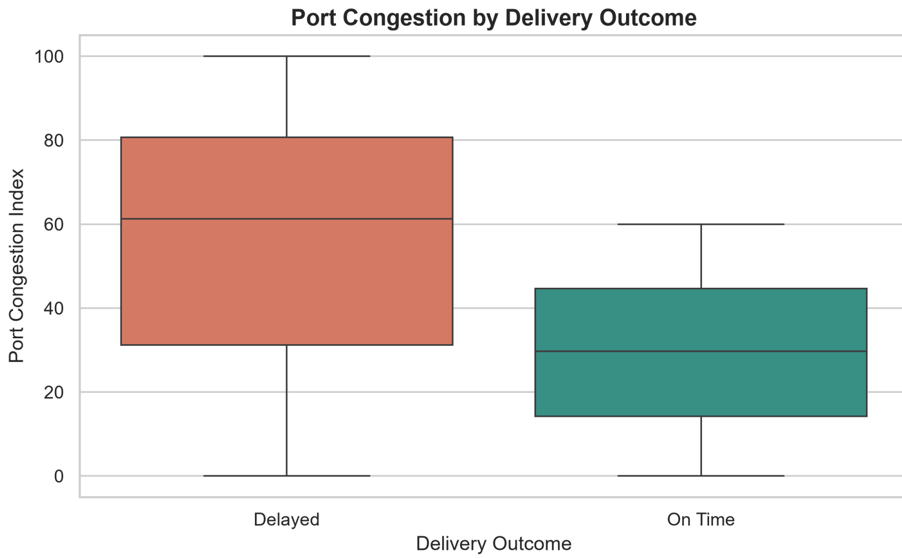

Fig. 4. Port congestion comparison between delayed and on-time deliveries.

Correlation analysis supported these findings. `Lead_Time_Days` had the strongest correlation with the delivery outcome at -0.589, followed by `Port_Congestion_Index` at -0.382. Since a value of 1 represents an on-time delivery, the negative correlations show that higher lead time and higher port congestion were associated with a lower chance of on-time delivery.

The correlation heatmap also showed that `Product_Price` and `Competitor_Price` were highly correlated at 0.993. `Carbon_Emission` and `Shipping_Cost` also had a strong correlation of 0.902. These variable pairs contained similar information and were considered during feature selection and model interpretation.

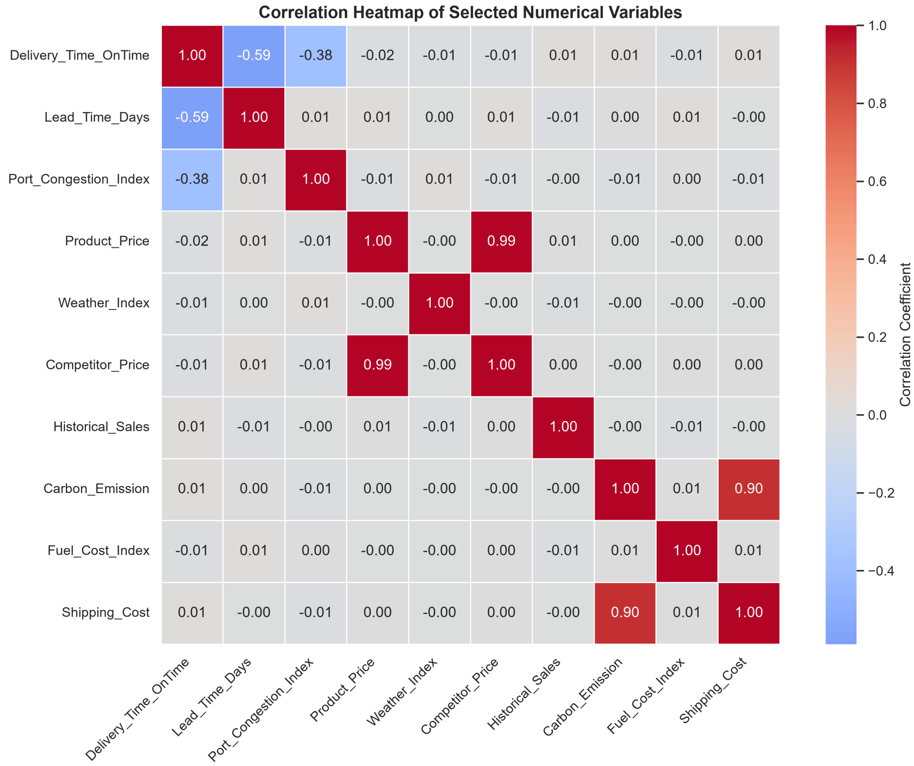

Fig. 5. Correlation heatmap of the numerical variables

Finally, the IQR method identified 1,175 possible outliers in `Carbon_Emission`, representing 4.70% of the dataset, and 629 possible outliers in `Shipping_Cost`, representing 2.52%. These values were not removed because they may represent valid high-cost or high-emission supply chain situations.

5.4 Key Learnings

The exploratory analysis showed that understanding the target distribution and predictor behavior was essential before model development. The delivery outcome was imbalanced, with 77.11% delayed deliveries and 22.89% on-time deliveries. Therefore, model evaluation required measures beyond accuracy, including balanced accuracy, precision, recall, F1-score, F2-score, confusion matrices, ROC-AUC, and PR-AUC.

The strongest univariate relationships with delivery performance were observed for `Lead_Time_Days` and `Port_Congestion_Index`. However, `Lead_Time_Days` was later excluded from the final predictor set because its timing was uncertain and it could introduce data leakage. Port congestion remained an important predictor and was later identified as the strongest model driver through both Logistic Regression coefficients and SHAP analysis.

The analysis also identified highly correlated variable pairs and possible outliers. These values were not removed automatically because they may represent valid business conditions. Instead, the findings were considered during feature selection, model interpretation, and limitation analysis.

Overall, the EDA phase provided the evidence needed to define the modeling target, review candidate predictors, identify class imbalance, reduce leakage risk, and select appropriate evaluation methods for predictive analysis.

6 Predictive Analysis and Model Building

6.1 Predictive Pipeline

The predictive application was developed as a supervised binary classification workflow. The original delivery-status variable, `Delivery_Time_OnTime`, was recoded as `Delivery_Delayed`, where 1 represents a delayed shipment and 0 represents an on-time shipment. This made delayed delivery the positive class because it is the primary operational risk addressed by the project.

Candidate variables were reviewed for identifiers, redundancy, outcome information, optimization outputs, and possible data leakage. After this review, 32 predictors were selected, consisting of 25 numerical variables and 7 categorical variables. The identifier fields, original date field, target variables, post-outcome variables, and `Lead_Time_Days` were excluded from the final predictor set.

The modeling pipeline included target preparation, feature selection, stratified data splitting, preprocessing, model training, model comparison, threshold selection, final testing, explainability analysis, and model saving. Numerical predictors were processed using median imputation and standardization. Categorical predictors were processed using most-frequent imputation and one-hot encoding. The preprocessing transformer was fitted only on training data to reduce the risk of data leakage.

Target Preparation → Feature Selection → Stratified Split → Preprocessing → Model Training → Validation-Based Threshold Selection → Final Testing → Explainability → Model Saving

The 32 original predictors were transformed into 64 model-ready features after numerical preprocessing and categorical one-hot encoding.

6.2 Machine Learning Algorithms

Four classification approaches were selected to provide a meaningful comparison between a simple baseline model, an interpretable linear model, and two nonlinear ensemble models. The algorithms were chosen based on the binary classification objective, the imbalanced delivery outcome, the structured tabular format of the dataset, and the need to compare predictive performance with model interpretability. The selected methods are standard supervised-learning approaches for classification tasks [3].

- **Dummy Classifier:** The Dummy Classifier was selected as a baseline because delayed deliveries represented 77.11% of the dataset. A majority-class baseline was necessary to determine whether the trained models provided meaningful improvement beyond predicting most shipments as delayed. It also demonstrated why accuracy alone could be misleading for this imbalanced dataset.
- **Logistic Regression:** Logistic Regression was selected because the project involved binary classification and required interpretable probability estimates for delayed delivery. It supported balanced class weights, probability-threshold adjustment, coefficient analysis, and SHAP-based explanation. These characteristics made it suitable for an explainable decision-support application.
- **Random Forest:** Random Forest was selected to evaluate whether nonlinear relationships and interactions among operational, logistics, geographic, supplier, and economic variables could improve classification performance. It also provided a tree-based ensemble comparison with the more interpretable Logistic Regression model.
- **Gradient Boosting:** Gradient Boosting was selected because it can learn complex patterns by building trees sequentially and correcting errors made by earlier trees. It was included to test whether a boosting-based ensemble could provide stronger separation between delayed and on-time shipments in the structured tabular dataset.

Together, these four approaches allowed the project to compare majority-class baseline performance, interpretable probability-based classification, nonlinear tree ensembles, class-imbalance behavior, and operational usefulness. Logistic Regression was selected as the final model because it achieved the highest ROC-AUC and PR-AUC values among the trained models, provided a more practical balance between delayed and on-time prediction, supported threshold optimization, and allowed clear interpretation through model coefficients and SHAP values.

6.3 Training, Validation, and Testing Process

The dataset contained 25,000 records. It was divided using an 80/20 stratified train-test split so that the delivery-class proportions remained similar in both datasets. The training dataset contained 20,000 records, while the untouched testing dataset contained 5,000 records.

The training dataset contained 15,422 delayed shipments and 4,578 on-time shipments. The testing dataset contained 3,856 delayed shipments and 1,144 on-time shipments. Stratification was important because the target variable was imbalanced, and preserving class proportions across data splits helps support more reliable evaluation of imbalanced classification problems [6].

For threshold selection, the original training dataset was divided again into 15,000 model-training records and 5,000 validation records. The validation dataset was used to compare probability thresholds without using the final testing data. The untouched test dataset was used only after the model and threshold-selection process had been completed.

The default classification threshold was 0.50. However, because missing a delayed shipment may be more costly than generating an additional delay alert, the validation process also evaluated alternative thresholds using the delayed-class F2-score. The F2-score gives more importance to recall than precision.

The highest balanced accuracy occurred at a threshold of 0.68, but this threshold detected only 51.21% of delayed shipments. A threshold of 0.37 was therefore selected as the business-oriented threshold because it increased delayed-shipment recall while maintaining an on-time recall of approximately 50%.

6.4 Implementation and Evaluation

The predictive analysis was implemented in Python using a Jupyter Notebook. Pandas and NumPy were used for data processing, while scikit-learn was used for preprocessing, model training, prediction, and evaluation. Matplotlib and Seaborn were used for visualizations, SHAP was used for explainability, and Joblib was used to save the final model package.

The models were evaluated using accuracy, balanced accuracy, delayed-class precision, delayed-class recall, F1-score, F2-score, ROC-AUC, PR-AUC, classification reports, and confusion matrices following standard classification-evaluation practices [4].

Accuracy was not used by itself because the majority-class Dummy Classifier could achieve 77.12% accuracy by predicting every record as delayed.

Balanced accuracy was used to evaluate performance across both classes. Precision measured how many predicted delays were actually delayed, while recall measured how many actual delays were detected. ROC-AUC evaluated ranking performance across classification thresholds, and PR-AUC was especially useful because the target classes were imbalanced.

The complete project source code, notebooks, saved model artifacts, dashboard implementation, and supporting project files are maintained in the project GitHub repository [7].

6.5 Model Comparison Results

Table 2 presents the performance of the four evaluated classification approaches using the unseen testing dataset.

Table 2. Comparison of Classification Model Performance

Model	Accuracy	Balanced Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Dummy Classifier	0.7712	0.5000	0.7712	1.0000	0.8708	0.5000	0.7712
Logistic Regression	0.6802	0.6940	0.8893	0.6686	0.7633	0.7671	0.9312
Random Forest	0.7692	0.5002	0.7713	0.9961	0.8694	0.7624	0.9280
Gradient Boosting	0.6288	0.7575	0.9970	0.5202	0.6837	0.7665	0.9306

The Dummy Classifier and Random Forest achieved relatively high accuracy because they predicted nearly every shipment as delayed. Their balanced accuracy values were close to 0.50, showing that they did not distinguish the two classes effectively.

Gradient Boosting achieved the highest balanced accuracy and correctly recognized nearly all on-time shipments. However, its delayed-shipment recall was only 52.02%, resulting in 1,850 missed delays. Logistic Regression provided a more useful balance and achieved the highest ROC-AUC of 0.7671 and PR-AUC of 0.9312.

6.6 Final Threshold Results

Table 3 compares the default Logistic Regression threshold of 0.50 with the selected business-oriented threshold of 0.37 on the final testing dataset.

Table 3. Logistic Regression Threshold Comparison on Test Data

Threshold	Accuracy	Balanced Accuracy	Precision	Recall	F1-Score	F2-Score	On-Time Recall
Default 0.50	0.6802	0.6940	0.8893	0.6686	0.7633	0.7035	0.7194
Business 0.37	0.7114	0.6386	0.8401	0.7728	0.8051	0.7854	0.5044

At the selected threshold of 0.37, the final Logistic Regression model correctly identified 2,980 delayed shipments and 577 on-time shipments. It missed 876 delayed shipments and produced 567 false delay alerts.

Compared with the default threshold, the business threshold reduced missed delays from 1,278 to 876. This represents 402 additional delayed shipments detected. The trade-off was an increase in false delay alerts and a reduction in on-time recall. This threshold was selected because the project gives greater importance to identifying shipments that may require early review.

6.7 Model Explainability and Saving

Logistic Regression coefficients and SHAP analysis were used to explain the final model [5]. Both methods identified `Port_Congestion_Index` as the strongest predictor of delayed-delivery risk. Its mean absolute SHAP value was 1.1072, substantially higher than the other predictors. Product price, competitor price, month, week, and shipping cost were among the next most influential variables.

The strong influence of port congestion should be interpreted cautiously because the dataset is synthetic. The relationship may partly reflect assumptions used when the dataset was generated and should not automatically be generalized to real operational supply chains without independent validation.

The final preprocessing transformer, trained Logistic Regression model, selected threshold, original predictor list, and transformed feature names were saved together using Joblib. The saved package was reloaded and tested successfully on sample records, confirming that predictions could be generated without retraining the model.

7 Project Methodology

The project followed a structured data analytics and machine learning workflow. The process began with defining the research problem and identifying delivery performance as the main outcome of interest. The selected public dataset was then reviewed to understand its structure, target variable, predictor attributes, data types, and overall suitability for the project.

The dataset was validated and cleaned by checking missing values, hidden placeholders, duplicate records, repeated identifiers, categorical inconsistencies, numeric ranges, and date formatting. The original dataset was preserved, while a separate cleaned version was created for analysis.

Exploratory data analysis was performed using descriptive statistics, frequency tables, bar charts, histograms, boxplots, grouped comparisons, correlation analysis, heatmaps, and the interquartile range method. This stage identified class imbalance, strong relationships involving lead time and port congestion, highly correlated predictor pairs, and possible outliers.

For predictive modeling, the delivery outcome was recoded as `Delivery_Delayed`, where 1 represents a delayed shipment and 0 represents an on-time shipment. Candidate variables were reviewed for identifiers, redundancy, timing uncertainty, post-outcome information, and possible data leakage. A total of 32 predictors were retained for the final machine learning workflow.

The dataset was divided using a stratified 80/20 train-test split. Numerical predictors were processed using median imputation and standardization, while categorical predictors were processed using most-frequent imputation and one-hot encoding. A Dummy Classifier, Logistic Regression, Random Forest, and Gradient Boosting model were trained and evaluated.

Because the target variable was imbalanced, model performance was assessed using accuracy, balanced accuracy, precision, recall, F1-score, F2-score, ROC-AUC, PR-AUC, classification reports, and confusion matrices. Logistic Regression was selected as the final model based on its overall balance, ranking performance, and interpretability.

A separate validation dataset was used to compare probability thresholds. A business-oriented threshold of 0.37 was selected to improve delayed-shipment detection. The final model was then evaluated on the untouched test dataset.

Logistic Regression coefficients and SHAP analysis were used to explain the model. The preprocessing transformer, trained model, selected threshold, and feature information were saved together as a reusable model package.

The completed project workflow was:

Problem Definition → Data Collection → Data Cleaning → Exploratory Data Analysis → Feature Selection → Preprocessing → Model Training → Threshold Selection → Final Evaluation → Explainability → Model Saving

8 Approach and Scope

The scope of this project is limited to predicting delivery-delay risk using the variables available in the selected public cross-border e-commerce dataset. The final predictive workflow uses 32 selected predictors and a Logistic Regression model with a business-oriented probability threshold of 0.37.

Because the dataset is synthetic, the model results should be interpreted as statistical associations within the selected dataset rather than evidence of real-world causation. The model is intended as a decision-support prototype and would require validation using real operational shipment data before practical business use.

9 Interpretation of Results and Conclusions

9.1 Charts and Visualizations

The charts selected for the results interpretation phase were chosen to show the delivery-outcome distribution, differences between delayed and on-time shipments, model performance, final classification outcomes, and the variables that influenced model predictions. The visualizations used in this project include the delivery-outcome bar chart, lead-time boxplot, port-congestion boxplot, correlation heatmap, model-performance comparison chart, final Logistic Regression confusion matrix, and SHAP feature-importance chart. Bar charts were appropriate for comparing class counts and model metrics, boxplots were useful for comparing numerical distributions between delivery outcomes, the correlation heatmap showed relationships among numerical variables, the confusion matrix summarized correct and incorrect classifications, and the SHAP chart explained the relative influence of the predictors.

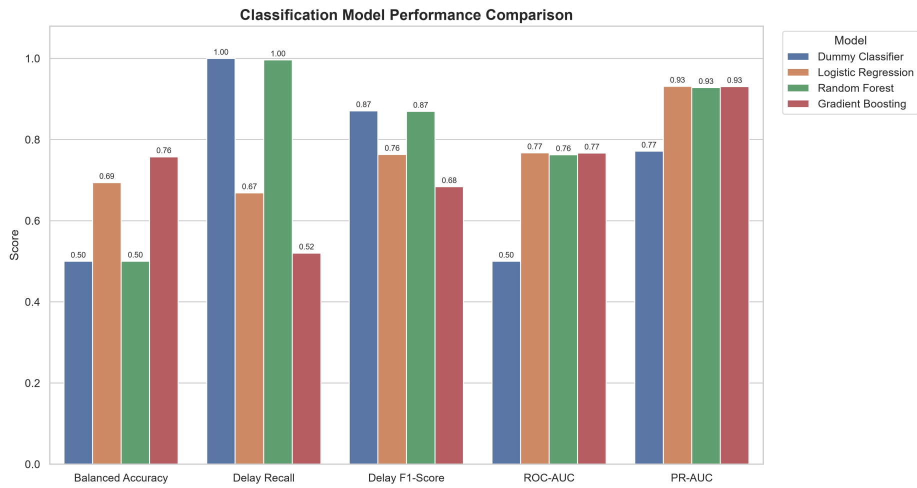

Fig. 6. Comparison of classification model performance.

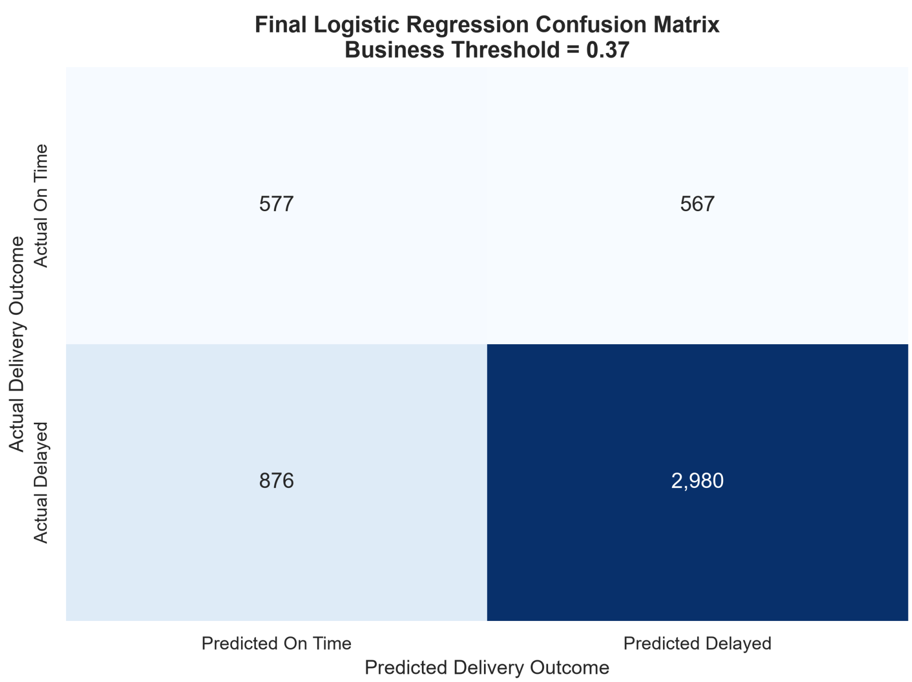

Fig. 7. Confusion matrix for the final Logistic Regression model at the 0.37 threshold

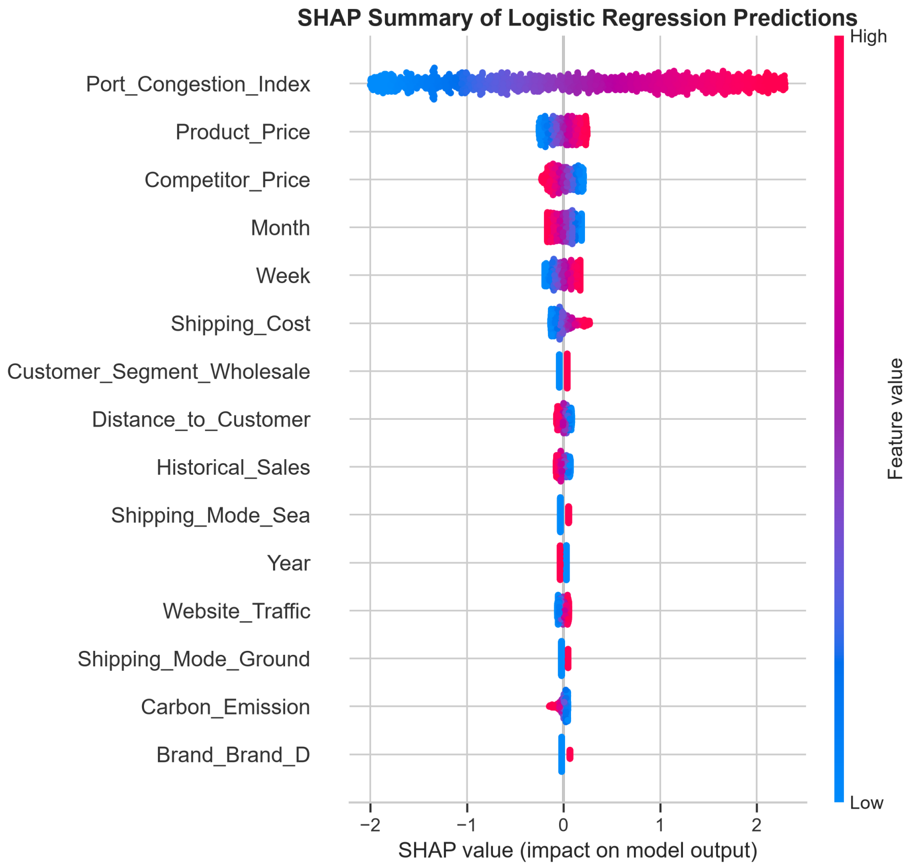

Fig. 8. SHAP feature importance for the final Logistic Regression model.

9.2 Results Explanation

The delivery-outcome chart in Fig. 2 showed that the dataset was imbalanced. Of the 25,000 records, 19,278 shipments, or 77.11%, were delayed, while 5,722 shipments, or 22.89%, were delivered on time. The lead-time boxplot in Fig. 3 showed a substantial difference between the two delivery groups. Delayed shipments had an average lead time of 21.42 days, compared with 8.54 days for on-time shipments. The port-congestion boxplot in Fig. 4 showed a similar pattern. Delayed shipments had an average Port Congestion Index of 55.89, while on-time shipments had an average value of 29.58.

The model-performance chart in Fig. 6 showed why accuracy alone was not sufficient for selecting the final model. The Dummy Classifier achieved 77.12% accuracy by predicting every shipment as delayed, but its balanced accuracy was only 0.5000. Random Forest produced a similar result, with 76.92% accuracy and 0.5002 balanced accuracy, because it classified nearly all records as delayed. Gradient Boosting achieved the highest balanced accuracy of 0.7575 and delayed-class precision of 0.9970, but its delayed-delivery recall was only 0.5202. This meant that it missed 1,850 actual delayed shipments. Logistic Regression achieved a balanced accuracy of 0.6940, delayed-class precision of 0.8893, delayed-class recall of 0.6686, ROC-AUC of 0.7671, and PR-AUC of 0.9312 at the default threshold. It provided the most practical balance between identifying delayed shipments, recognizing on-time shipments, ranking delivery risk, and supporting interpretation.

The final confusion matrix in Fig. 7 presents the Logistic Regression results after applying the selected probability threshold of 0.37. The model correctly identified 2,980 delayed shipments and 577 on-time shipments. It missed 876 delayed shipments and produced 567 false delay alerts. Compared with the default threshold of 0.50, the selected threshold reduced missed delays from 1,278 to 876, allowing the model to detect 402 additional delayed shipments. Delayed-delivery recall increased from 0.6686 to 0.7728, while the F1-score increased from 0.7633 to 0.8051 and the F2-score increased from 0.7035 to 0.7854. The trade-off was a reduction in on-time recall from 0.7194 to 0.5044 and an increase in false delay alerts.

The SHAP feature-importance chart in Fig. 8 showed that Port Congestion Index was the strongest driver of the final Logistic Regression predictions, with a mean absolute SHAP value of 1.1072. Its influence was substantially greater than the other predictors. Product price, competitor price, month, week, and shipping cost were among the next most influential features, but their SHAP values were much lower. This result indicates that port congestion played the largest role in separating delayed and on-time shipment risk within the fitted model.

9.3 Data Inference

The analysis showed that delayed deliveries were common throughout the dataset rather than being limited to one product category, shipping mode, region, or country. The delivery percentages across these categorical groups were relatively similar, which suggests that no single categorical variable was sufficient to explain delivery performance by itself. Instead, delivery risk appeared to result from a combination of operational, logistics, geographic, economic, and market-related factors.

Lead time and port congestion showed the clearest differences between delayed and on-time shipments during exploratory analysis. Longer lead times were strongly associated with delayed outcomes, while higher port-congestion values were also associated with a greater likelihood of delay. However, Lead Time Days was excluded from the final predictive model because its timing was uncertain and it could introduce data leakage. Port Congestion Index remained in the predictor set and was later identified as the strongest driver of model predictions.

The results also showed that class imbalance had a major effect on model behavior. Models that predicted nearly every shipment as delayed achieved relatively high overall accuracy but failed to distinguish on-time shipments. This confirmed that accuracy alone was not an appropriate measure for evaluating the predictive application. Balanced accuracy, recall, precision, F1-score, F2-score, ROC-AUC, PR-AUC, and confusion-matrix results were required to understand the practical value of each model.

The threshold analysis further showed that model performance depends on the operational objective. Lowering the Logistic Regression threshold from 0.50 to 0.37 increased the number of delayed shipments detected, but it also produced more false delay alerts. Therefore, the selected threshold represents a business trade-off between reducing missed delays and avoiding unnecessary intervention.

9.4 Statistical Conclusions

The statistical evidence showed that the delivery outcome was imbalanced, with 77.11% delayed shipments and 22.89% on-time shipments. This imbalance explained why the Dummy Classifier achieved 77.12% accuracy while producing a balanced accuracy of only 0.5000. The Random Forest model showed a similar pattern, with 76.92% accuracy and 0.5002 balanced accuracy. These results confirmed that high overall accuracy did not necessarily represent useful class separation.

Logistic Regression achieved the highest ROC-AUC value of 0.7671 and the highest PR-AUC value of 0.9312 among the evaluated models. These scores showed that the model had a useful ability to rank delayed shipments above on-time shipments across probability thresholds. At the selected threshold of 0.37, the model achieved 71.14% accuracy, 63.86% balanced accuracy, 84.01% delayed-delivery precision, 77.28% delayed-delivery recall, an F1-score of 0.8051, and an F2-score of 0.7854.

The final confusion matrix provided additional statistical evidence. Of the 3,856 delayed shipments in the test dataset, 2,980 were identified correctly and 876 were missed. Of the 1,144 on-time shipments, 577 were classified correctly and 567 were incorrectly flagged as delayed. Compared with the default threshold of 0.50, the selected threshold identified 402 additional delayed shipments.

Correlation analysis also supported the observed delivery patterns. Lead Time Days had a correlation of -0.589 with the original on-time delivery outcome, while Port Congestion Index had a correlation of -0.382. Because the original target used 1 for on-time delivery, these negative correlations indicated that higher lead time and higher port congestion were associated with a lower probability of on-time delivery. These results represent statistical associations and should not be interpreted as proof of causation.

9.5 General Observations

One important observation was that the model with the highest overall accuracy was not necessarily the most useful model for the project. The Dummy Classifier and Random Forest achieved relatively high accuracy because delayed shipments represented the majority class, but both models performed poorly when evaluated across delayed and on-time deliveries together. This reinforced the importance of selecting evaluation measures that match the structure of the data and the business objective.

Another observation was that the probability threshold had a direct effect on the operational usefulness of the model. The default threshold of 0.50 produced fewer false delay alerts, but it missed more delayed shipments. The selected threshold of 0.37 increased delayed-shipment detection and reduced missed delays, although it required accepting more false alerts. This means that the final threshold should be reviewed periodically based on the cost of missed delays, the cost of operational intervention, and the available review capacity.

The explainability results also showed the importance of combining predictive performance with model interpretation. Port Congestion Index was identified as the strongest predictor by both Logistic Regression coefficient analysis and SHAP. However, this result should be interpreted carefully because the dataset is synthetic. The identified relationships may partly reflect the assumptions used to generate the data and should be validated using real operational shipment records before business deployment.

The final model is therefore best treated as a decision-support tool rather than an automatic decision-making system. It can help prioritize shipments that may require early review, but final operational actions should also consider current logistics conditions, business rules, expert judgment, and independently validated data. The interactive dashboard developed as part of the broader capstone project provides a practical way to communicate predicted delivery risk and support further analysis.

9.6 Conclusion

The analysis demonstrated that supply chain, logistics, market, geographic, and operational variables can provide useful information for predicting delivery-delay risk in the selected cross-border e-commerce dataset. Logistic Regression was selected as the final model because it provided the most practical combination of delayed-shipment detection, on-time recognition, ranking performance, threshold flexibility, and interpretability.

The business-oriented threshold of 0.37 improved delayed-delivery recall from 66.86% to 77.28% and detected 402 additional delayed shipments compared with the default threshold of 0.50. Port Congestion Index was the strongest predictor in both the Logistic Regression coefficient analysis and SHAP interpretation. However, because the dataset is synthetic, these findings should be treated as statistical and model-based associations rather than evidence of real-world causation.

Overall, the final model provides an interpretable foundation for identifying shipments that may require earlier review. Its predictions should support, rather than replace, operational judgment. Validation using real shipment data would be required before the model could be used in a production supply chain environment.

10 Deployment, Limitations, and Future Work

The completed capstone includes the full data analytics and machine learning workflow from data collection and cleaning through exploratory analysis, predictive modeling, threshold selection, explainability, model packaging, and interactive decision-support implementation.

A PyShiny application was developed to operationalize the final Logistic Regression model. The application loads the saved preprocessing pipeline and trained model, applies the selected decision threshold of 0.37, and allows users to evaluate delivery-delay risk under user-defined supply chain conditions. The application was also deployed as a publicly accessible web application using shinyapps.io:

`https://kiruthikaa2512.shinyapps.io/cb-supply-risk/`

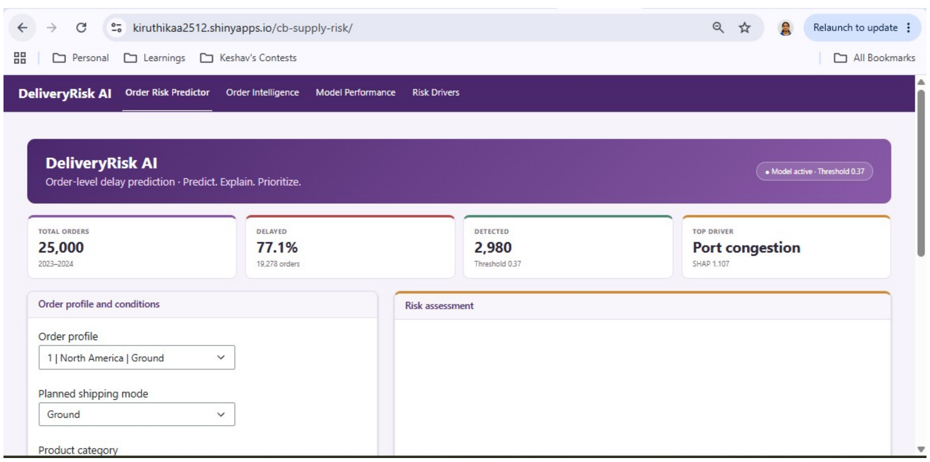

Fig. 9. Production PyShiny dashboard overview showing the interactive delivery-risk decision-support interface.

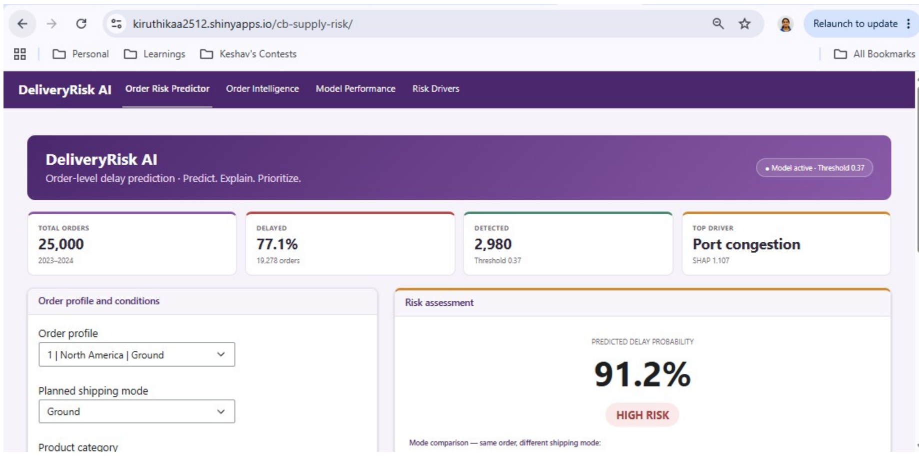

Fig. 10. Interactive delivery-risk prediction view showing the model output for a user-defined supply chain scenario.

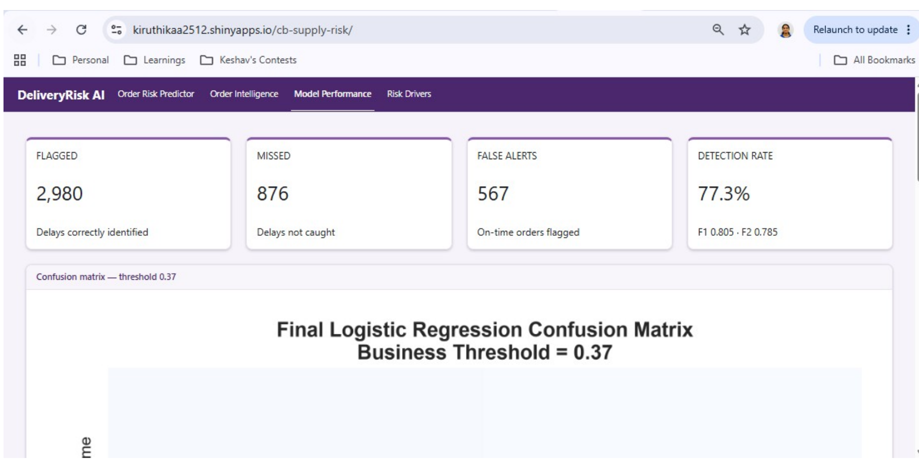

Fig. 11. Dashboard model-performance view summarizing final classification outcomes and the confusion matrix at the selected 0.37 threshold.

The dashboard demonstrates how the predictive workflow can be translated into an interactive decision-support tool. It is intended as a prototype rather than a production operational system and is not connected to live carrier, warehouse, weather, port, supplier, or enterprise data sources.

The primary limitation of the project is that the dataset is synthetic. Therefore, the relationships identified by the models may partly reflect the assumptions used during data generation and should not be interpreted as evidence of real-world causation. Before operational use, the workflow should be validated using real shipment records and current supply chain conditions.

Future work could include integrating live operational data, monitoring model performance and data drift, periodically recalibrating the decision threshold, evaluating additional predictive algorithms, and incorporating current external risk information such as weather, port congestion, geopolitical events, and transportation disruptions.

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